Name: ____________________

ALGEBRAIC EXPRESSIONS, (SUBSTITUTION INTO EXPRESSIONS)

Date: ________________

LO: I can substitute given values into algebraic expressions and evaluate the result.

What does 'substitute' mean?

Substitution means replacing a **letter** with a given number, then working out the value of the expression.

Remember: 2x means 2 × x. When we write letters next to numbers or other letters, multiplication is implied.

Quick examples

●	If x = 3, then 2x = 6	2 × 3
●	If a = 4, then a + 7 = 11	4 + 7
●	If x = 5, then x^2 = 25	5 × 5
●	If x = 3, then 2x = 23	not joining digits
●	If x = 4, then x^2 = 8	squared means × itself, not × 2

Key idea

Replace the letter with the number. Use BIDMAS to work out the answer in the correct order.

If x = 4: 3x + 1 = 3 × 4 + 1 = 13

Substitute then calculate

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – single variable

Find the value of 4x + 3 when x = 5

Replace x with 5: 4 × 5 + 3

Work out the multiplication: 20 + 3

Add: 23

= 23

Multiply first (BIDMAS), then add.

Example 2 – two variables

Find the value of 3a + 2b when a = 4 and b = 5

Replace a with 4 and b with 5: 3 × 4 + 2 × 5

Work out each multiplication: 12 + 10

Add: 22

= 22

Substitute both letters, then follow BIDMAS.

Example 3 – with powers

Find the value of x^2 + 3x when x = 4

Replace x with 4: 4^2 + 3 × 4

Work out the power: 16 + 3 × 4

Multiply then add: 16 + 12 = 28

= 28

Powers first, then multiplication, then addition.

Example 4 – negative values

Find the value of $2x^2$ - 3x when x = -2

Replace x with -2: 2 × $(-2)^2$ - 3 × (-2)

Work out $(-2)^2$ = 4, so 2 × 4 = 8

Work out -3 × (-2) = 6, so 8 + 6 = 14

= 14

$(-2)^2$ = 4, not -4. A negative squared gives a positive.

YOUR TURN – PRACTICE (deliberate practice)

1. Single variable, no powers

Substitute $x = 3$ into each expression.

a) $2x + 1$

b) $5x - 4$

c) $10 - 2x$

d) $4x + 7$

2. Two variables

Substitute $a = 5$ and $b = 2$ into each expression.

a) $3a + b$

b) $4a - 2b$

c) $a + 3b$

d) $2a - b + 1$

3. Expressions with powers

Substitute $x = 3$ into each expression.

a) x^2

b) $x^2 + 5$

c) $2x^2$

d) $x^2 + 2x$

4. Negative substitution

Substitute $x = -2$ into each expression.

a) $3x + 10$

b) x^2

c) $x^2 + x$

d) $5 - 2x$

5. Mixed challenge

Use the values given with each question.

a) $3x - 2y$ when $x = 4, y = 3$

b) $a^2 + b^2$ when $a = 3, b = 4$

c) $2x^2 - x + 1$ when $x = 5$

d) $4(x + y)$ when $x = 3, y = -1$

e) $x^2 - y^2$ when $x = 5, y = 3$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) If $x = 3$, then $2x = 23$ ☐

Correct answer: __________

Reason: __________________

b) If $x = 4$, then $x^2 = 16$ ☐

Correct answer: __________

Reason: __________________

c) If $x = -3$, then $x^2 = -9$ ☐

Correct answer: __________

Reason: __________________

d) If $a = 2$ and $b = 3$, then $ab = 6$ ☐

Correct answer: __________

Reason: __________________

e) If $x = 5$, then $3x + 1 = 16$ ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The formula for the area of a trapezium is $A = \frac{1}{2}(a + b)h$. Find the area when **a = 5 cm**, **b = 9 cm** and **h = 4 cm**. Show your working.

Expression: ____________________________

Simplified: ____________________________

b) A taxi firm charges **£(2.50 + 1.20m)** for a journey of **m miles**. Work out the cost for a journey of **8 miles**, then find how many miles you can travel for exactly **£20.50**.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!